

***BiT
FMIE
DIMENSIONING***

BY: KALKIDAN M.

Dimensioning

- ✓ A drawing of a component, in addition to providing complete shape description, must also furnish information regarding the size description.
- ✓ These are provided through the distances between the surfaces, location of holes, nature of surface finish, type of material, etc.
- ✓ The expression of these features on a drawing, using lines, symbols, figures and notes is called dimensioning.

General principles:

- Dimension is a numerical value expressed in appropriate units of measurement and indicated on drawings, using lines, symbols, notes, etc., so that all features are completely defined.
- 1- As far as possible, dimensions should be placed outside the view.
 - 2- Dimensions should be taken from visible outlines rather than from hidden lines.
 - 3- Dimensioning to a center line should be avoided except when the center line passes through the center of a hole.

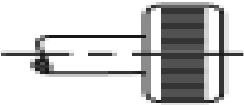
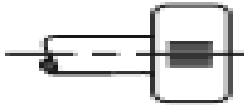
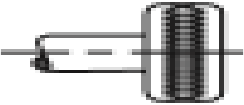
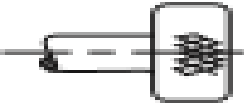
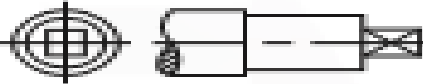
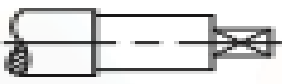
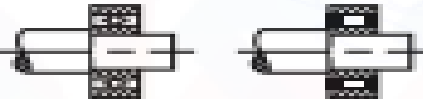
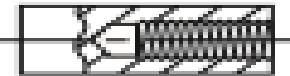
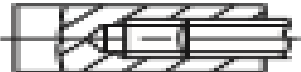
4- Each feature should be dimensioned once only on a drawing.

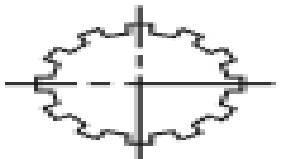
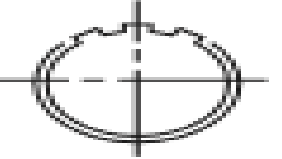
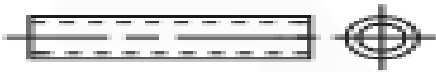
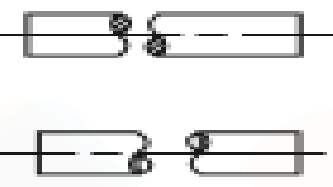
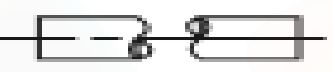
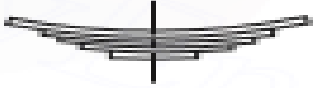
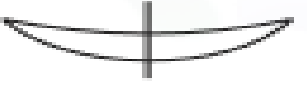
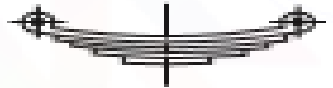
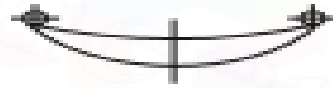
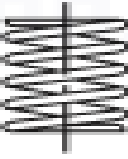
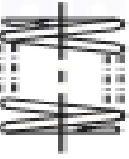
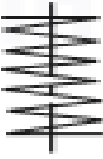
5- Dimensions should be placed on the view or section that relates most clearly to the corresponding features.

6- Each drawing should use the same unit for all dimensions, but without showing the unit symbol.

7- No more dimensions than are necessary to define a part should be shown on a drawing.

8- No features of a part should be defined by more than one dimension in any one direction.

Title	Subject	Convention
Straight knurling		
Diamond knurling		
Square on shaft		
Holes on circular pitch		
Bearings		
External screw threads (Detail)		
Internal screw threads (Detail)		
Screw threads (Assembly)		

Title	subject		Convention
Splined shafts			
Interrupted views	 		 
Semi-elliptic leaf spring			
Semi-elliptic leaf spring with eyes			
	Subject	Convention	Diagrammatic Representation
Cylindrical compression spring			
Cylindrical tension spring			

Title	Convention
Spur gear	
Bevel gear	
Worm wheel	
Worm	

Fig. 2.27 Conventional representation of machine components

Methods of execution

- ✓ The elements of dimensioning include the projection line, dimension line, leader line, dimension line termination, the origin indication and the dimension itself.

The following are some of the principles to be adopted during execution of dimensioning:

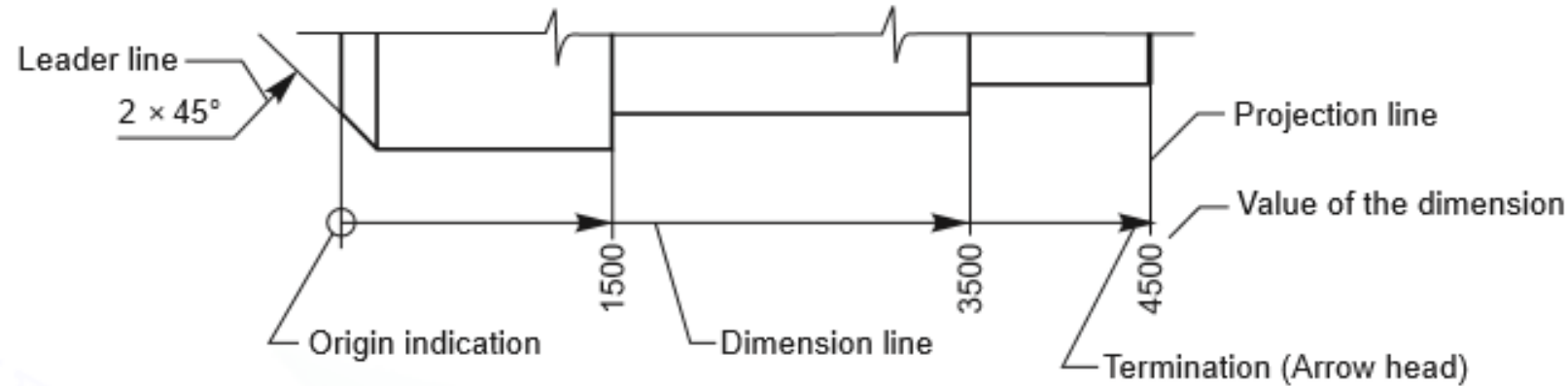


Fig. 2.28 Elements of dimensioning

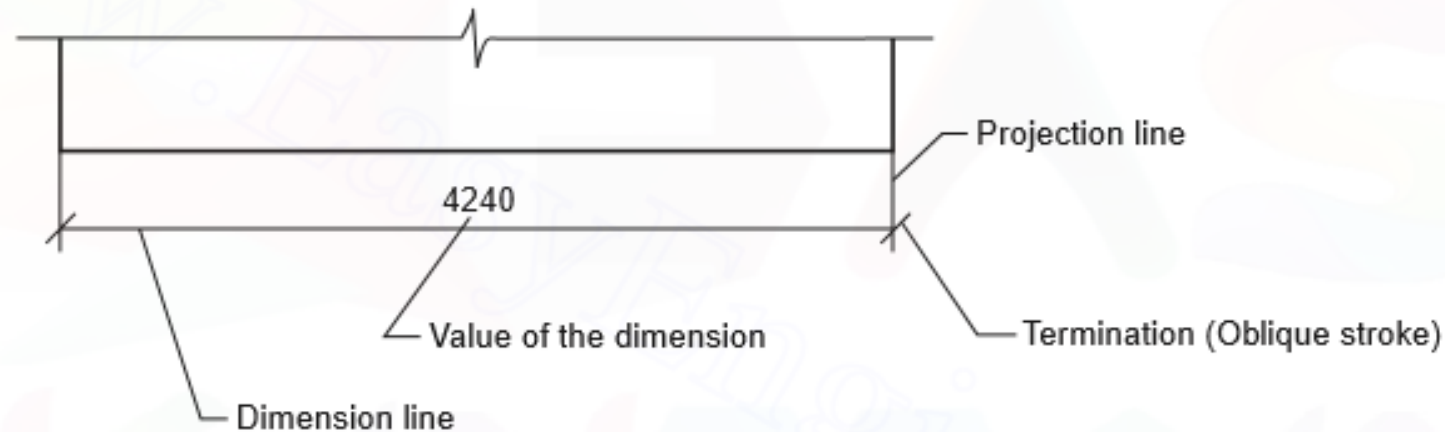


Fig. 2.29

1. Projection and dimension lines should be drawn as thin continuous lines.
2. Projection lines should extend slightly beyond the respective dimension lines.
3. Projection lines should be drawn perpendicular to the feature being dimensioned. Where necessary, they may be drawn obliquely, but parallel to each other (Fig. 2.30). However, they must be in contact with the feature.
4. Projection lines and dimension lines should not cross each other, unless it is unavoidable (Fig. 2.31).
5. A dimension line should be shown unbroken, even where the feature to which it refers, is shown broken (Fig. 2.32).
6. A center line or the outline of a part should not be used as a dimension line, but may be used in place of projection line (Fig. 2.31).

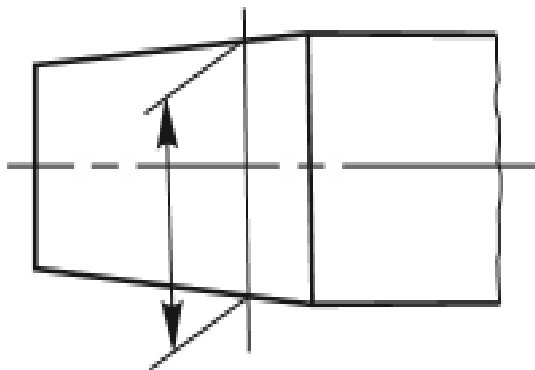


Fig. 2.30

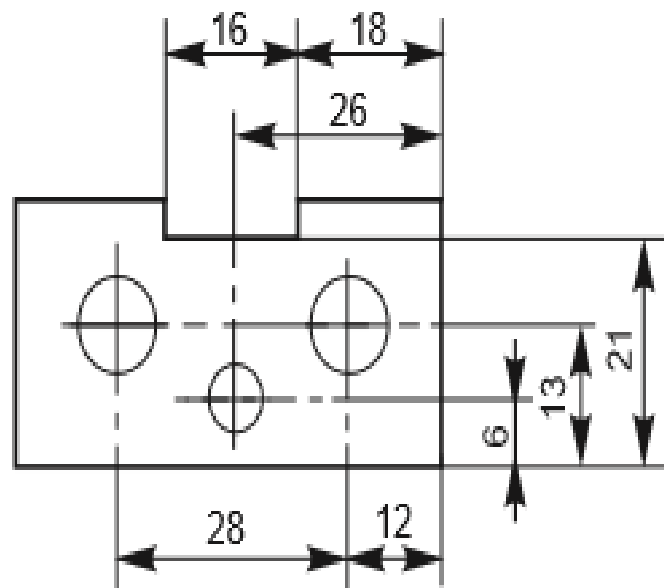


Fig. 2.31

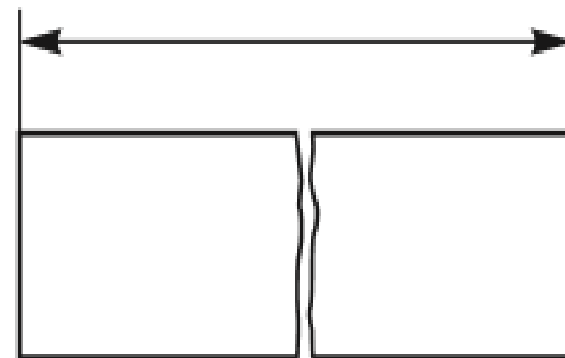


Fig. 2.32

Termination and origin indication

- Dimension lines should show distinct termination, in the form of arrow heads or oblique strokes or where applicable, an origin indication.
- Two dimension line terminations and an origin indication are shown in Fig. 2.33 below: In this,
 1. the arrow head is drawn as short lines, having an included angle of 15° , which is closed and filled-in.
 2. the oblique stroke is drawn as a short line, inclined at 45° .
 3. the origin indication is drawn as a small open circle of approximately 3 mm in diameter.

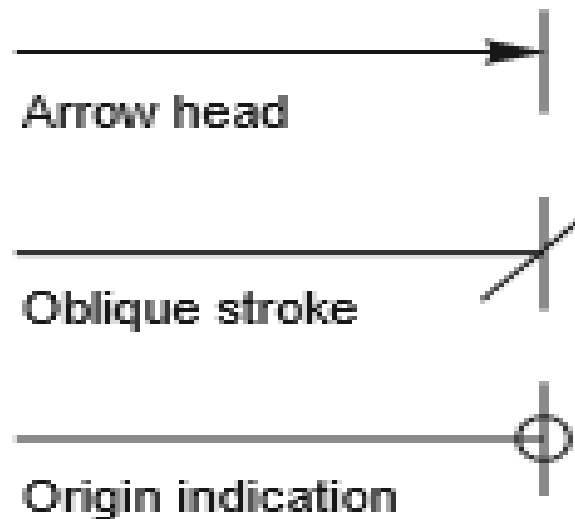


Fig. 2.33

- ✓ The size of the terminations should be proportionate to the size of the drawing on which they are used.
- ✓ Where space is limited, arrow head termination may be shown outside the intended limits of the dimension line that is extended for that purpose.
- ✓ In certain other cases, an oblique stroke or a dot may be substituted (Fig. 2.34).
- ✓ Where a radius is dimensioned, only one arrow head termination, with its point on the arc end of the dimension line, should be used (Fig. 2.35).
- ✓ However, the arrow head termination may be either on the inside or outside of the feature outline, depending upon the size of feature.

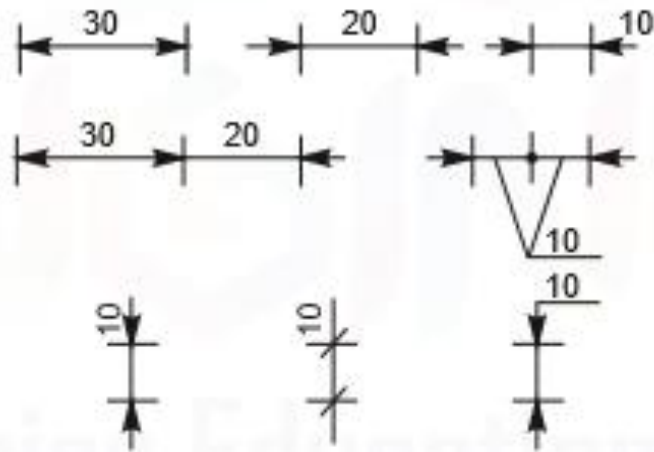


Fig. 2.34

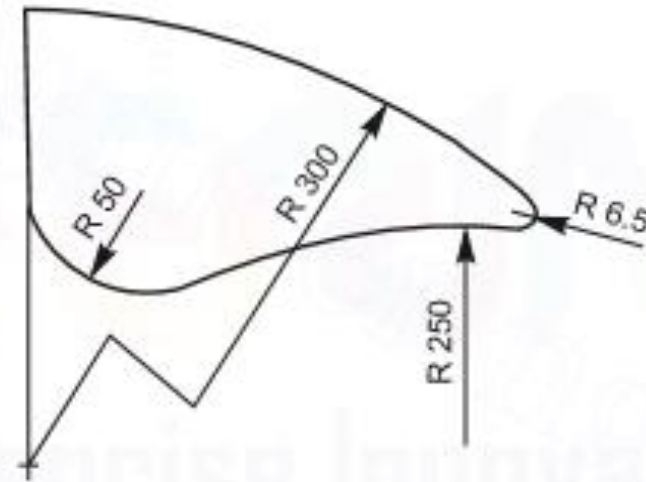


Fig. 2.35

Methods of Dimensioning:

- Dimensions should be shown on drawings in characters of sufficient size, to ensure complete legibility.
- They should be placed in such a way that they are not crossed or separated by any other line on the drawing.
- Dimensions should be indicated on a drawing, according to one of the following two methods. However, only one method should be used on any one drawing.

METHOD–1 (Aligned System)

- Dimensions should be placed parallel to their dimension lines and preferably near the middle, above and clear-off the dimension line (Fig. 2.36).
- An exception may be made where superimposed running dimensions are used (Fig. 2.44 b)
- Dimensions may be written so that they can be read from the bottom or from the right side of the drawing.
- Dimensions on oblique dimension lines should be oriented as shown in Fig. 2.37. Angular dimensions may be oriented as shown in Fig. 2.38.

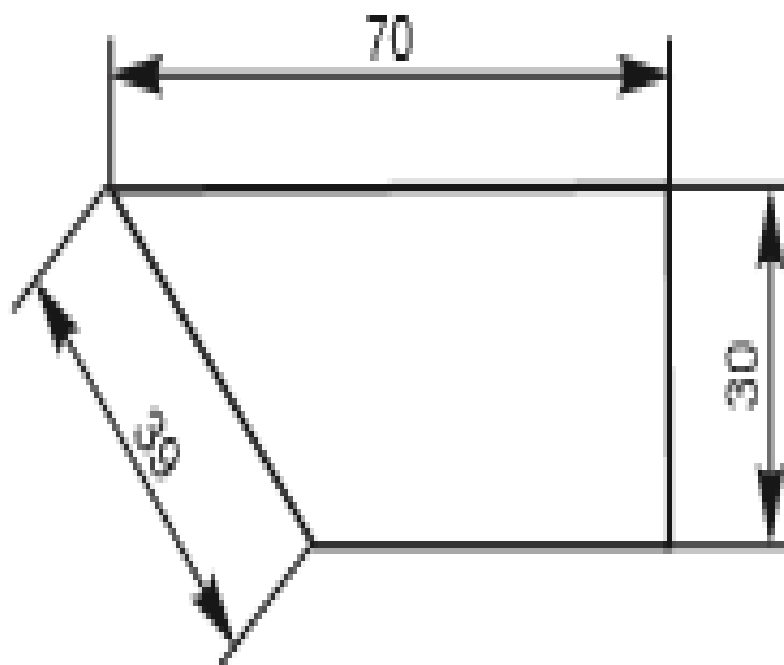


Fig. 2.36

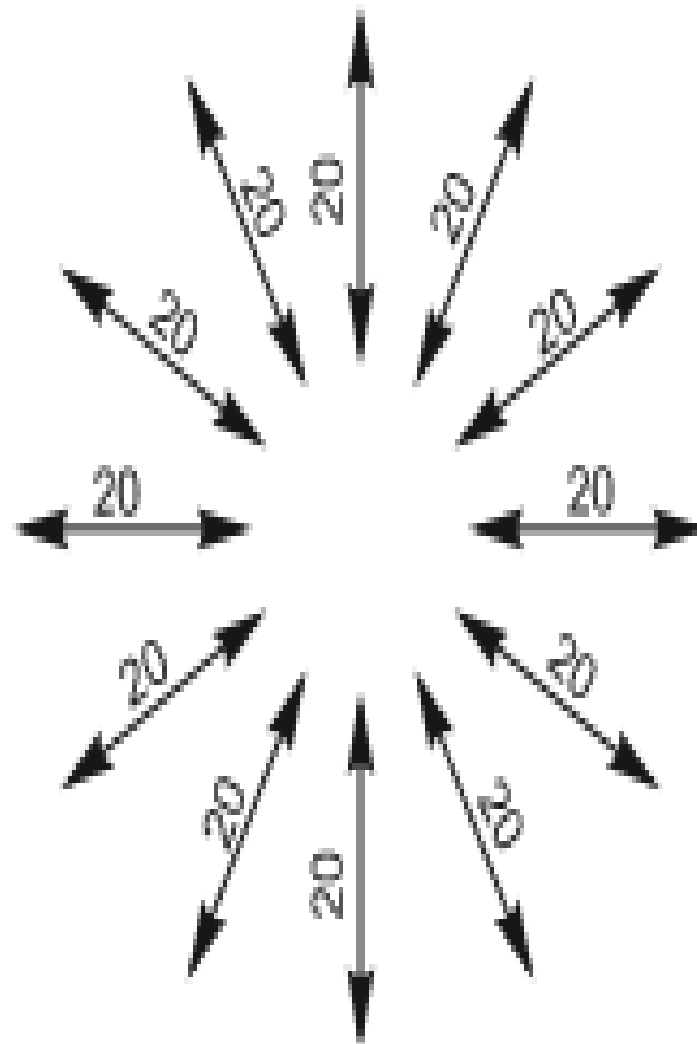


Fig. 2.37 Oblique dimensioning

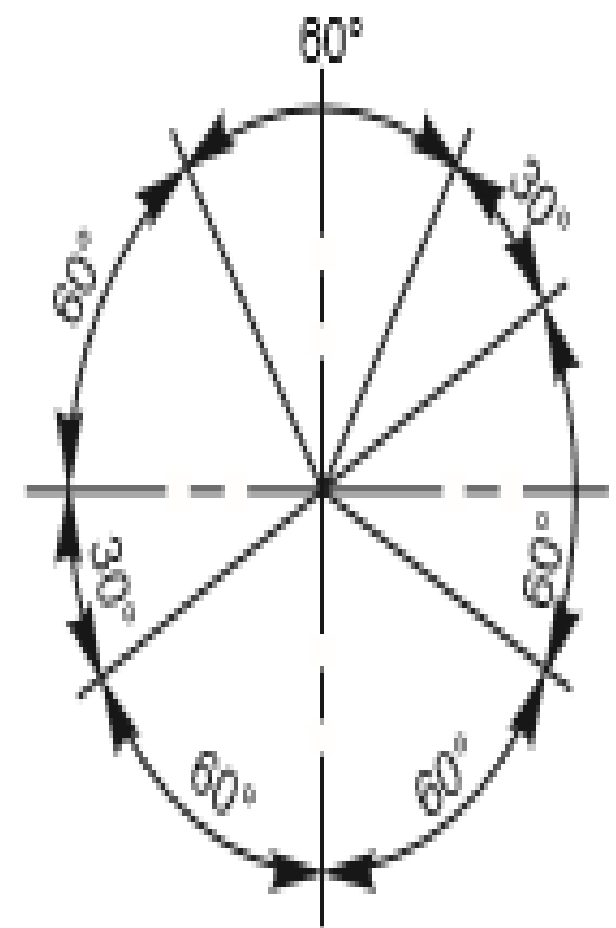


Fig. 2.38 Angular dimensioning

METHOD-2 (Uni-directional System)

Dimensions should be indicated so that they can be read from the bottom of the drawing only. Non-horizontal dimension lines are interrupted, preferably near the middle, for insertion of the dimension (Fig. 2.39).

Angular dimensions may be oriented as in Fig. 2.40.

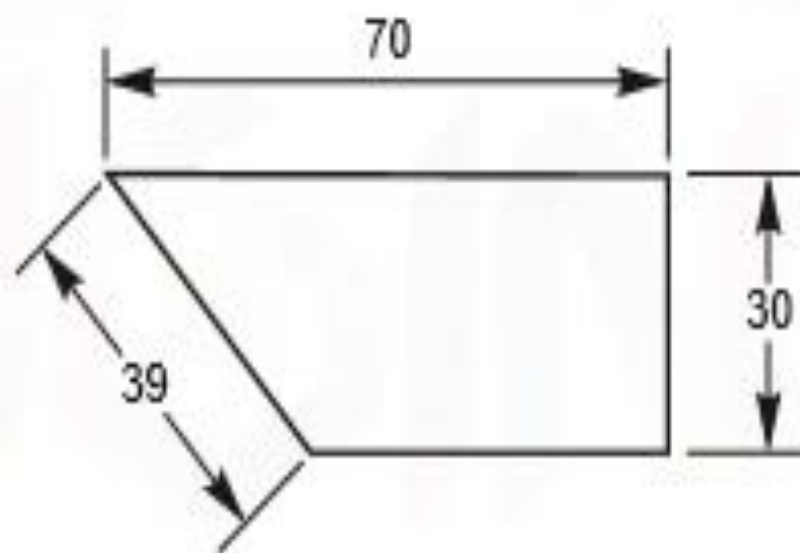


Fig. 2.39

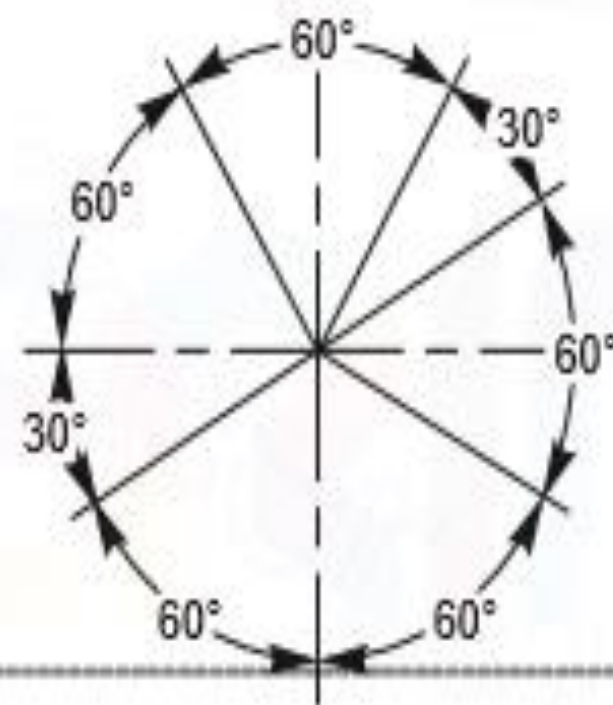


Fig. 2.40 Angular dimensioning

Dimensions can be, (i) above the extension of the dimension line, beyond one of the terminations, where space is limited (Fig. 2.34) or (ii) at the end of a leader line, which terminates on a dimension line, that is too short to permit normal dimension placement (Fig. 2.34) or (iii) above a horizontal extension of a dimension line, where space does not allow placement at the interruption of a non-horizontal dimension line (Fig. 2.41). Values of dimensions, out of scale (except where break lines are used) should be underlined as shown in Fig. 2.41.

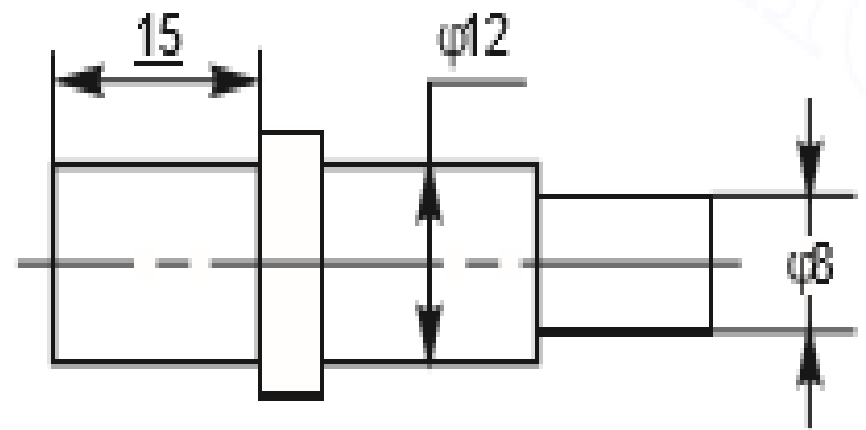


Fig. 2.41

The following indications (symbols) are used with dimensions to reveal the shape identification and to improve drawing interpretation. The symbol should precede the dimensions (Fig. 2.42).

ϕ : Diameter $S\phi$: Spherical diameter R : Radius SR : Spherical radius $\square$: Square

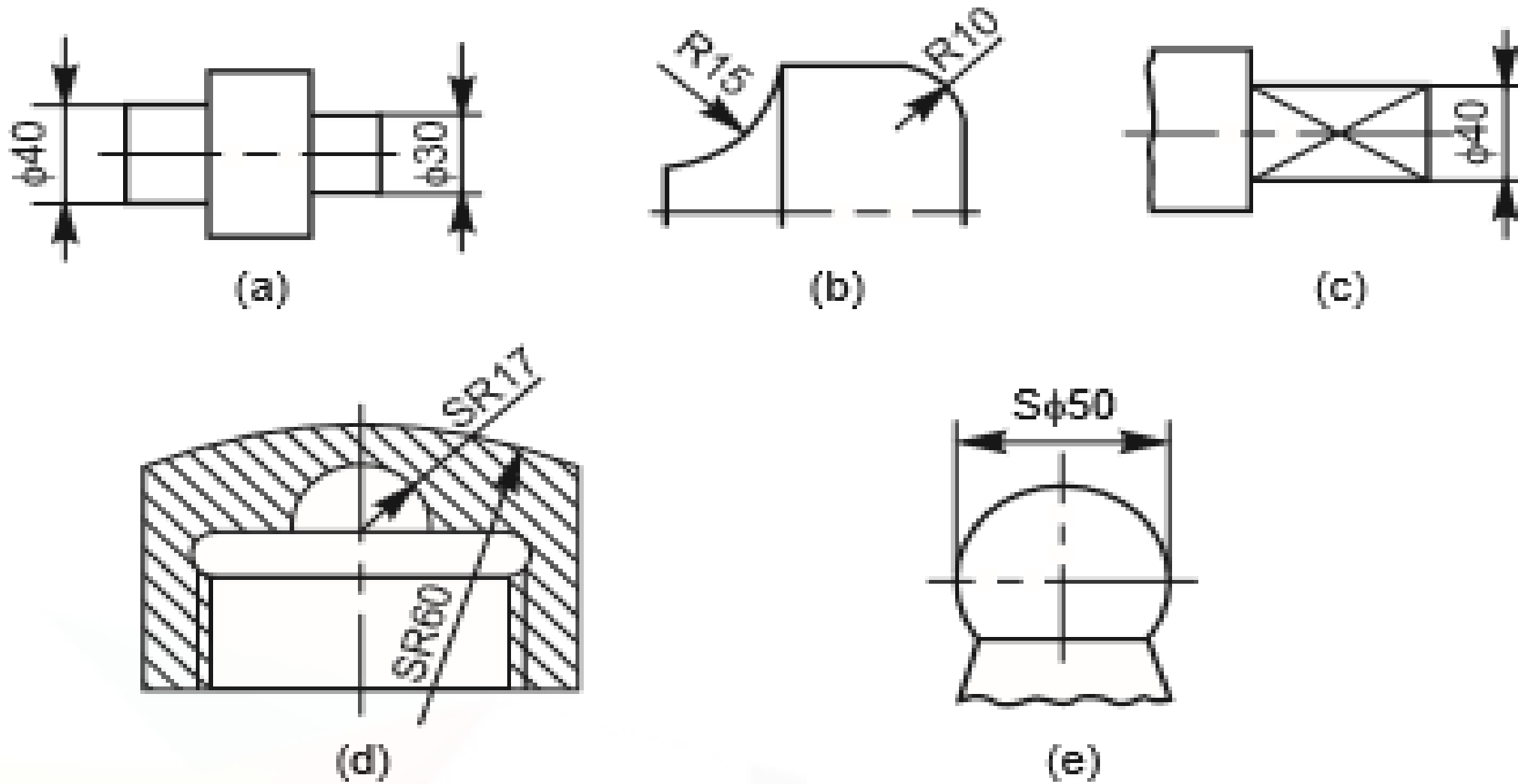


Fig. 2.42 Shape identification symbols

2.8.5 Arrangement of Dimensions

The arrangement of dimensions on a drawing must indicate clearly the design purpose. The following are the ways of arranging the dimensions.

2.8.5.1 Chain Dimensions

Chains of single dimensions should be used only where the possible accumulation of tolerances does not endanger the functional requirement of the part (Fig. 2.43).

2.8.5.2 Parallel Dimensions

In parallel dimensioning, a number of dimension lines, parallel to one another and spaced-out are used. This method is used where a number of dimensions have a common datum feature (Fig. 2.44 a).

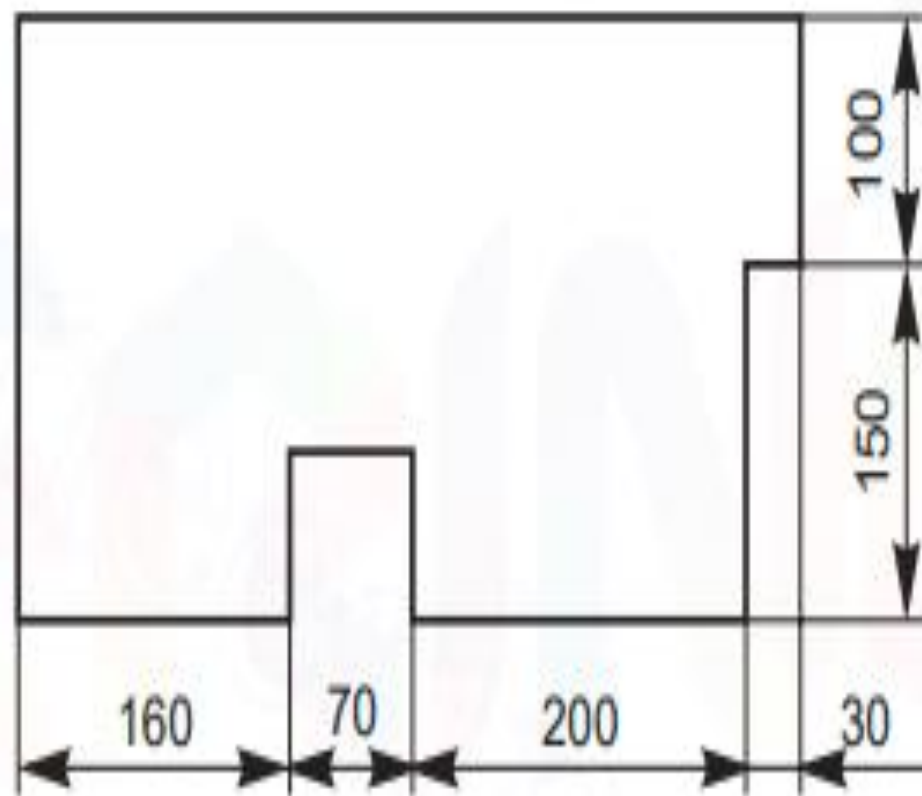
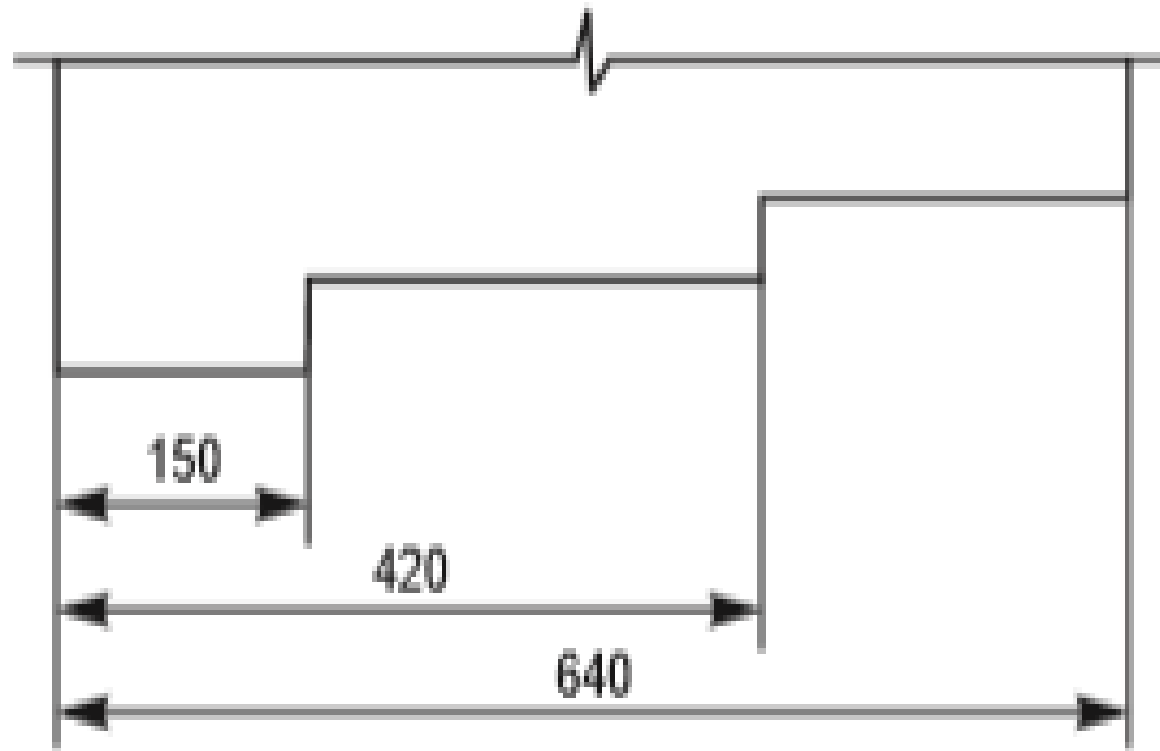


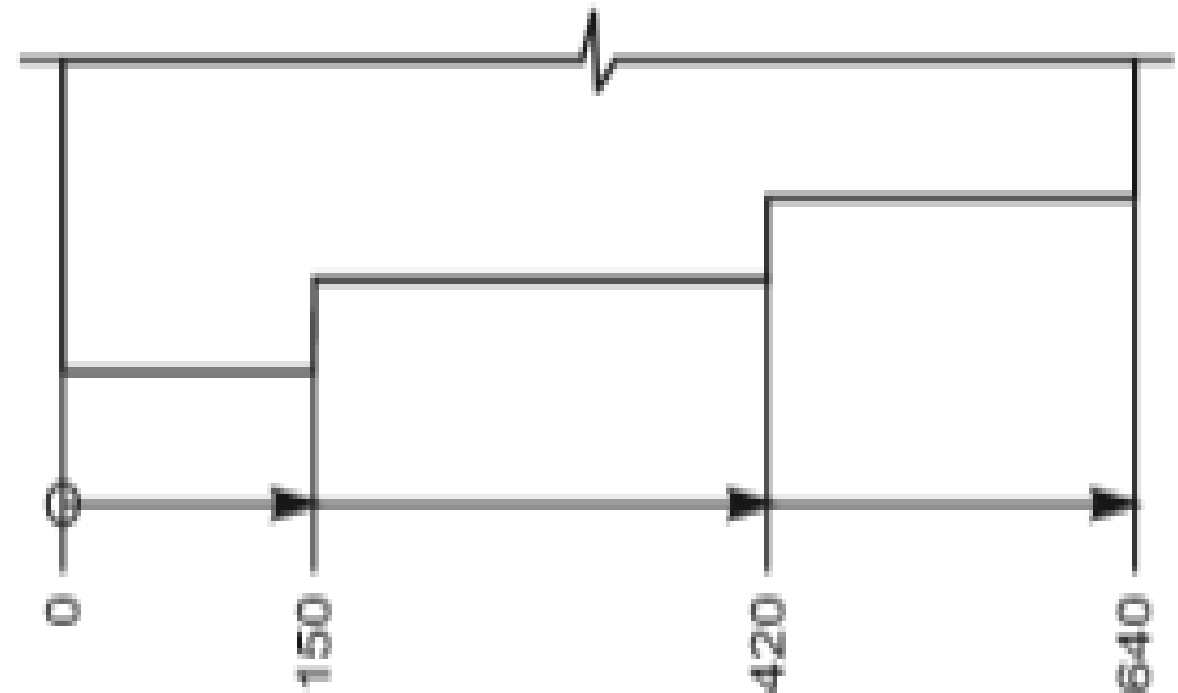
Fig. 2.43 Chain dimensioning

2.8.5.3 Super-imposed Running Dimensions

These are simplified parallel dimensions and may be used where there are space limitations (Fig. 2.44 b).



(a)



(b)

Fig. 2.44 Parallel dimensioning

2.8.5.4 Combined Dimensions

These are the result of simultaneous use of chain and parallel dimensions (Fig. 2.45).

2.8.5.5 Co-ordinate Dimensions

The sizes of the holes and their co-ordinates may be indicated directly on the drawing; or they may be conveniently presented in a tabular form, as shown in Fig. 2.46.

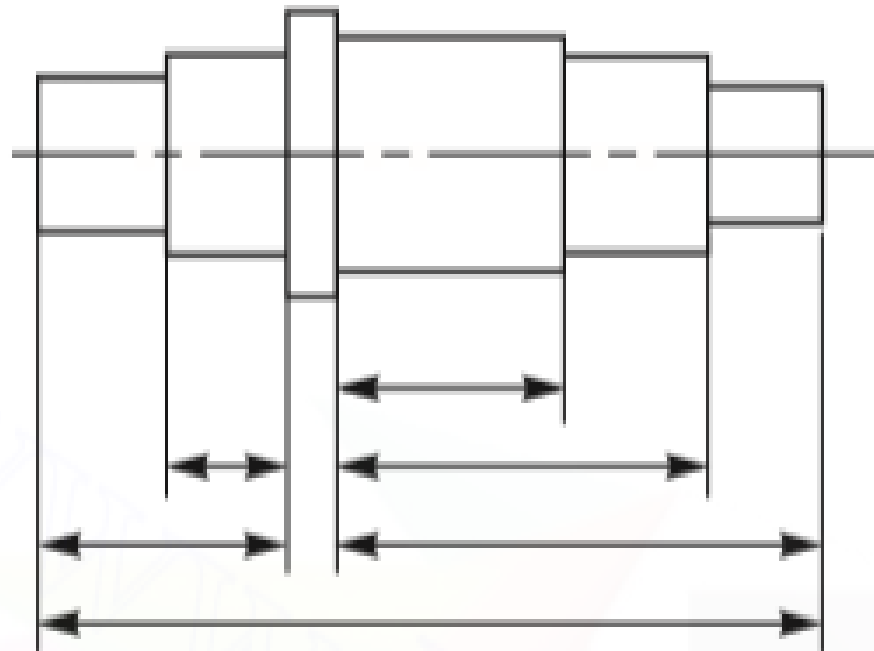


Fig. 2.45 Combined dimensioning

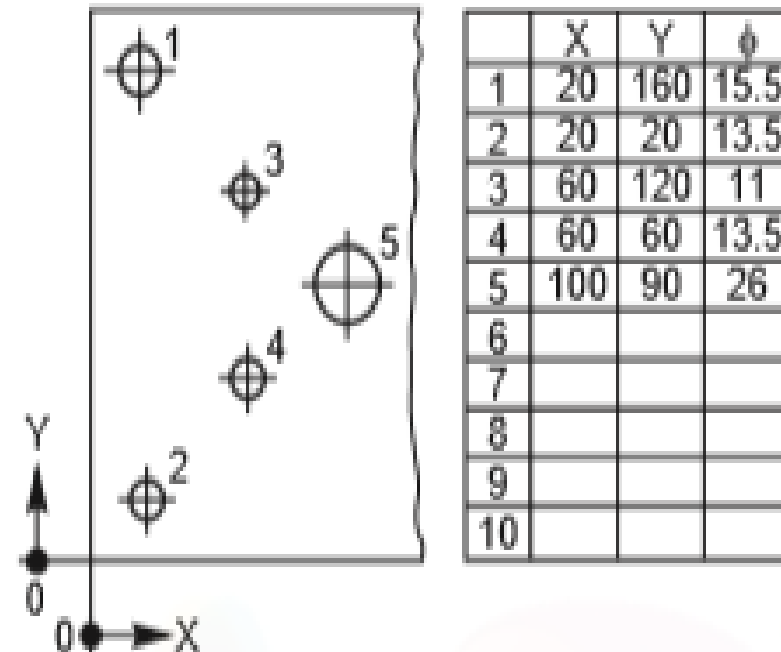


Fig. 2.46 Co-ordinate dimensioning

2.8.6 Special Indications

2.8.6.1 *Diameters*

Diameters should be dimensioned on the most appropriate view to ensure clarity. The dimension value should be preceded by ϕ . Figure 2.47 shows the method of dimensioning diameters.

2.8.6.2 *Chords, Arcs, Angles and Radii*

The dimensioning of chords, arcs and angles should be as shown in Fig. 2.48. Where the centre of an arc falls outside the limits of the space available, the dimension line of the radius should be broken or interrupted according to whether or not it is necessary to locate the centre (Fig. 2.35).

Where the size of the radius can be derived from other dimensions, it may be indicated by a radius arrow and the symbol R, without an indication of the value (Fig. 2.49).

2.8.6.3 Equi-distant Features

Linear spacings with equi-distant features may be dimensioned as shown in Fig. 2.50.

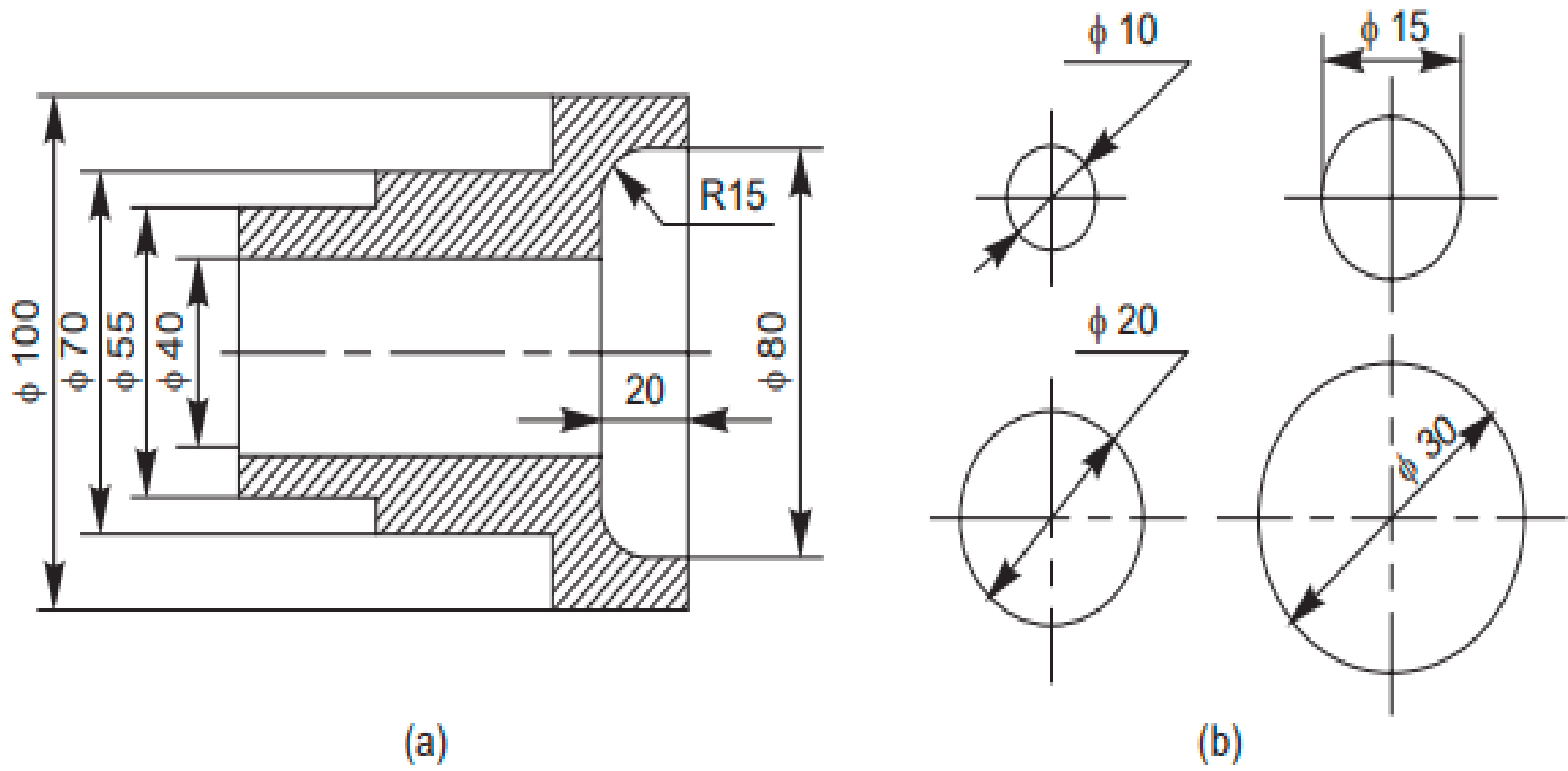


Fig. 2.47 Dimensioning of diameters

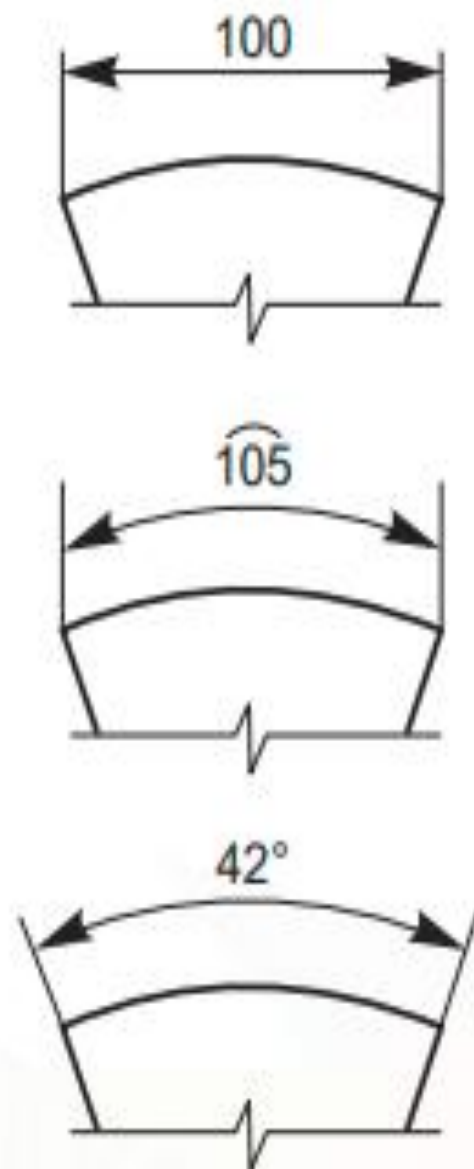


Fig. 2.48 Dimensioning of chords, arcs and angles

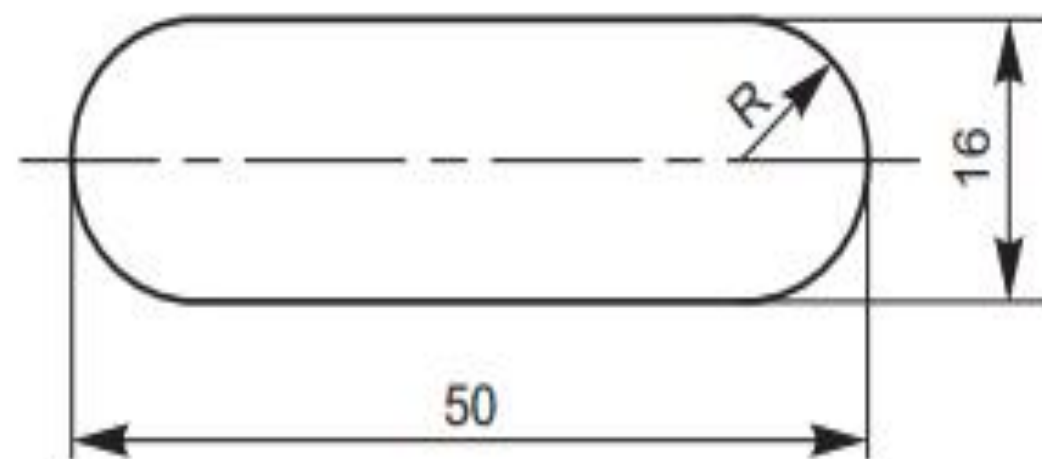


Fig. 2.49 Dimensioning of radius

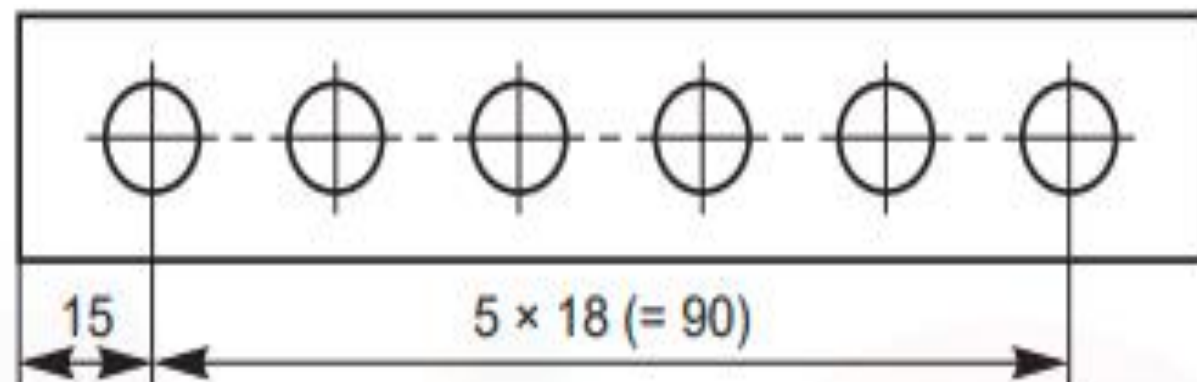


Fig. 2.50 Dimensioning equi-distant features

2.8.6.4 Chamfers and Countersunks

Chamfers may be dimensioned as shown in Fig. 2.51 and countersunks, as shown in Fig. 2.52.

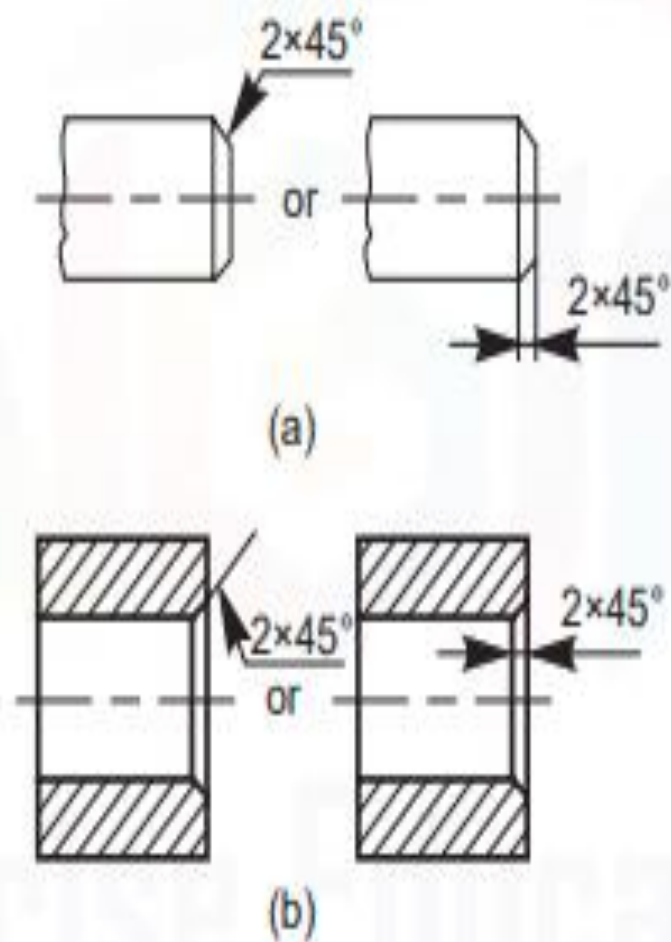


Fig. 2.51 Dimensioning chamfers

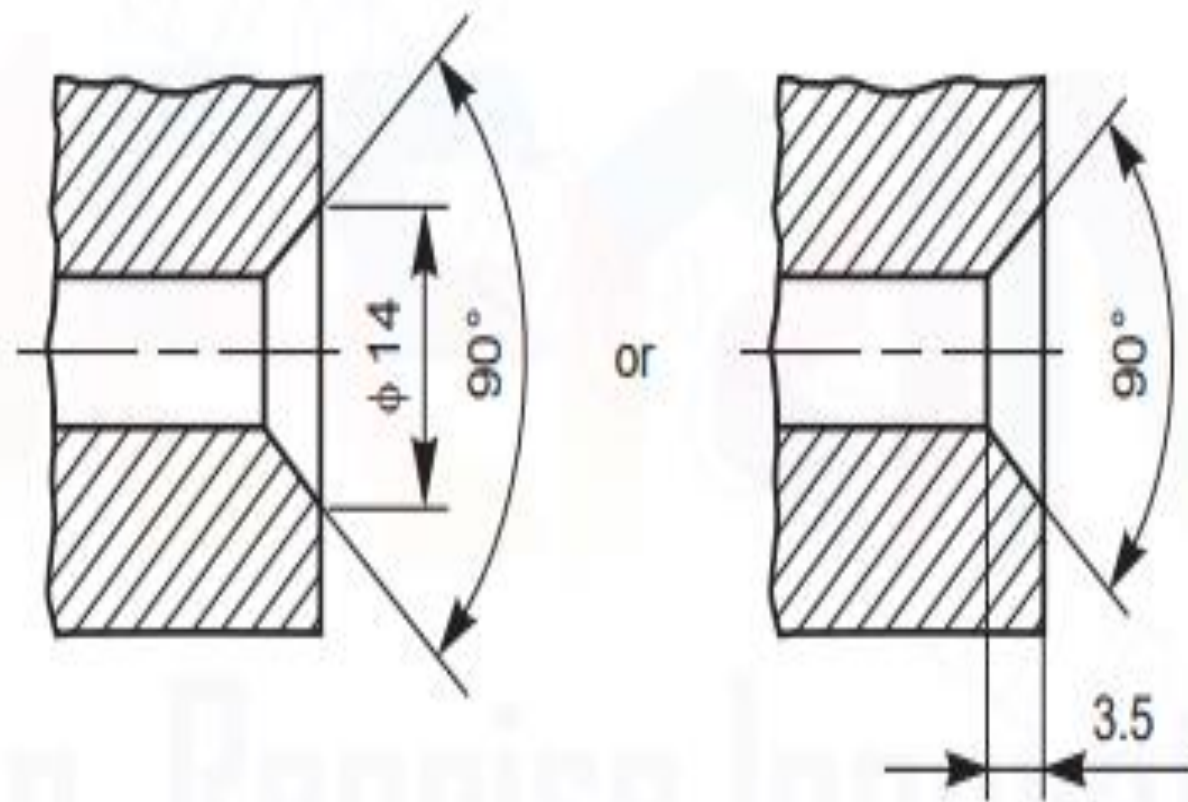


Fig. 2.52 Dimensioning countersunks

2.8.6.5 Screw Threads

Screw threads are always specified with proper designation. The nominal diameter is preceded by the letter M. The useful length of the threaded portion only should be dimensioned as shown in Fig. 2.53. While dimensioning the internal threads, the length of the drilled hole should also be dimensioned (Fig. 2.53).

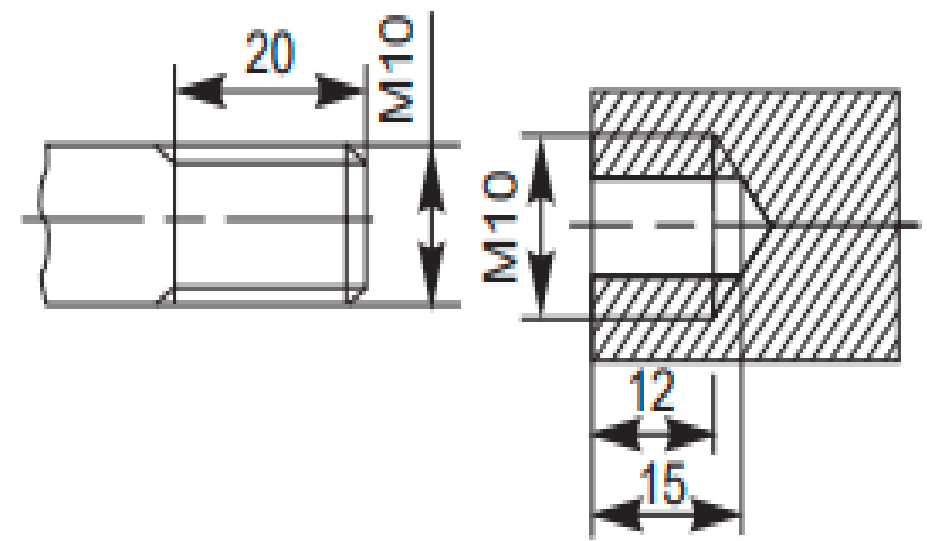


Fig. 2.53 Dimensioning screw threads

2.8.6.6 Tapered Features

Tapered features are dimensioned, either by specifying the diameters at either end and the length, or the length, one of the diameters and the taper or the taper angle (Fig. 2.54 *a*).

A slope or flat taper is defined as the rise per unit length and is dimensioned by the ratio of the difference between the heights to its length (Fig. 2.54 *b*).

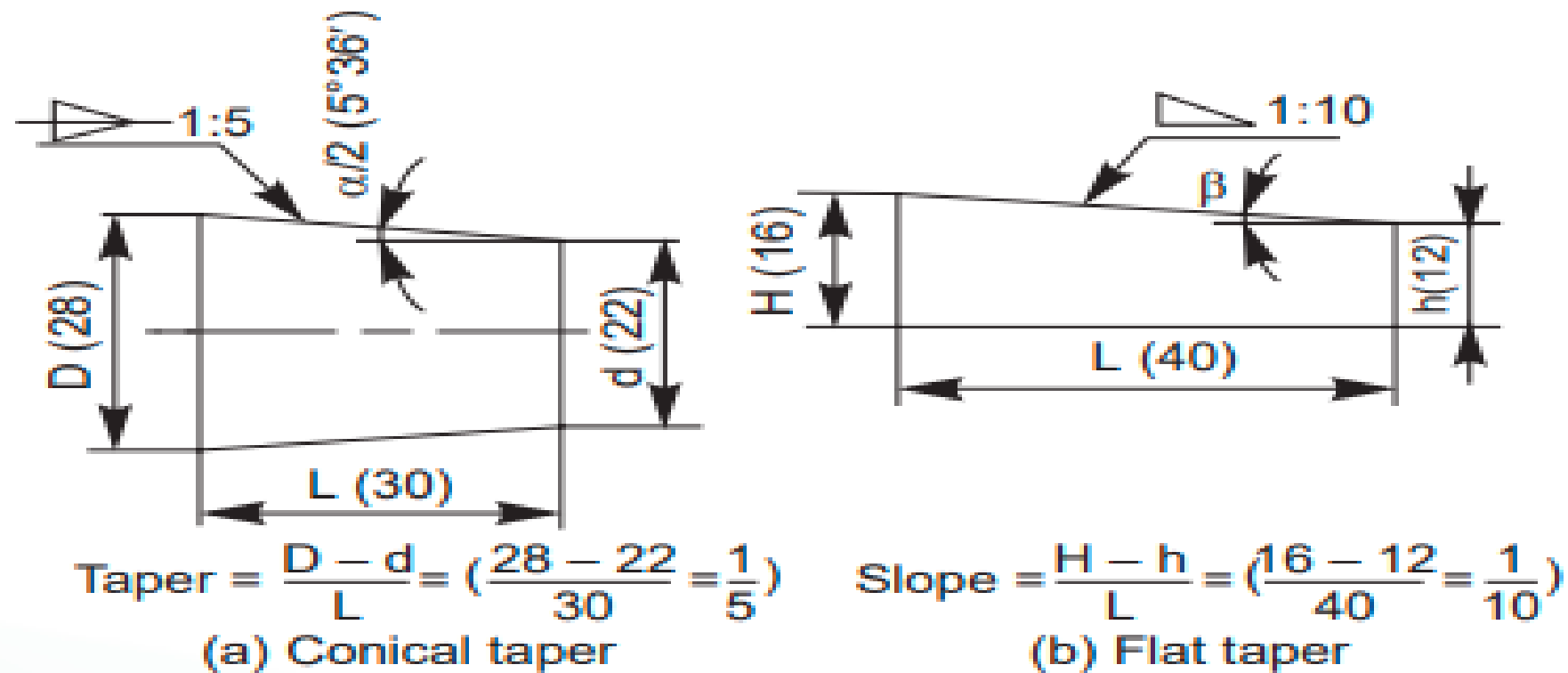
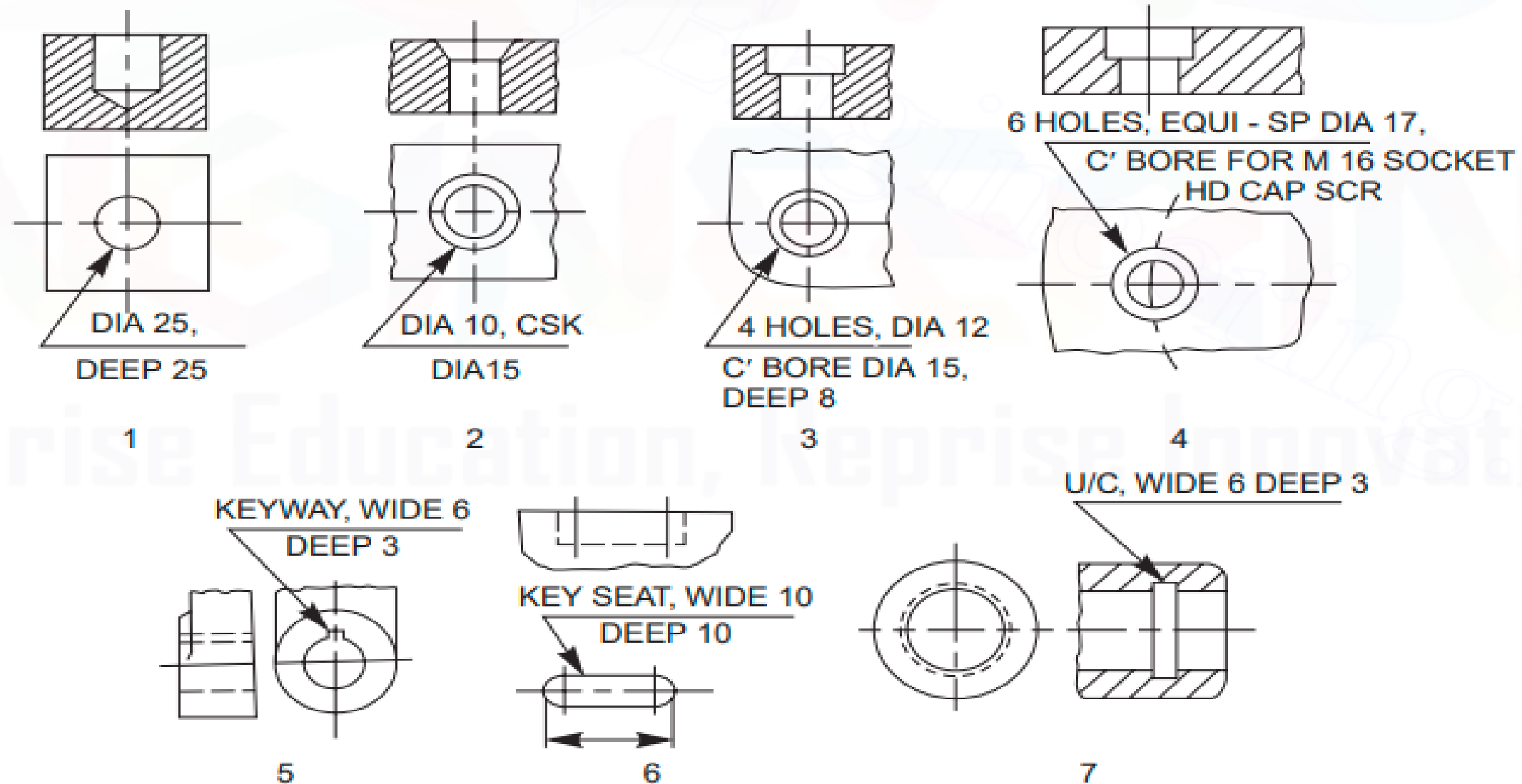


Fig. 2.54 Dimensioning tapered features

2.8.6.7 Notes

Notes should always be written horizontally in capital letters and begin above the leader line and may end below also. Further, notes should be brief and clear and the wording should be standard in form. The standard forms of notes and the method of indication, for typical cases is shown in Fig. 2.55. The meaning of the notes is given in Table 2.8.



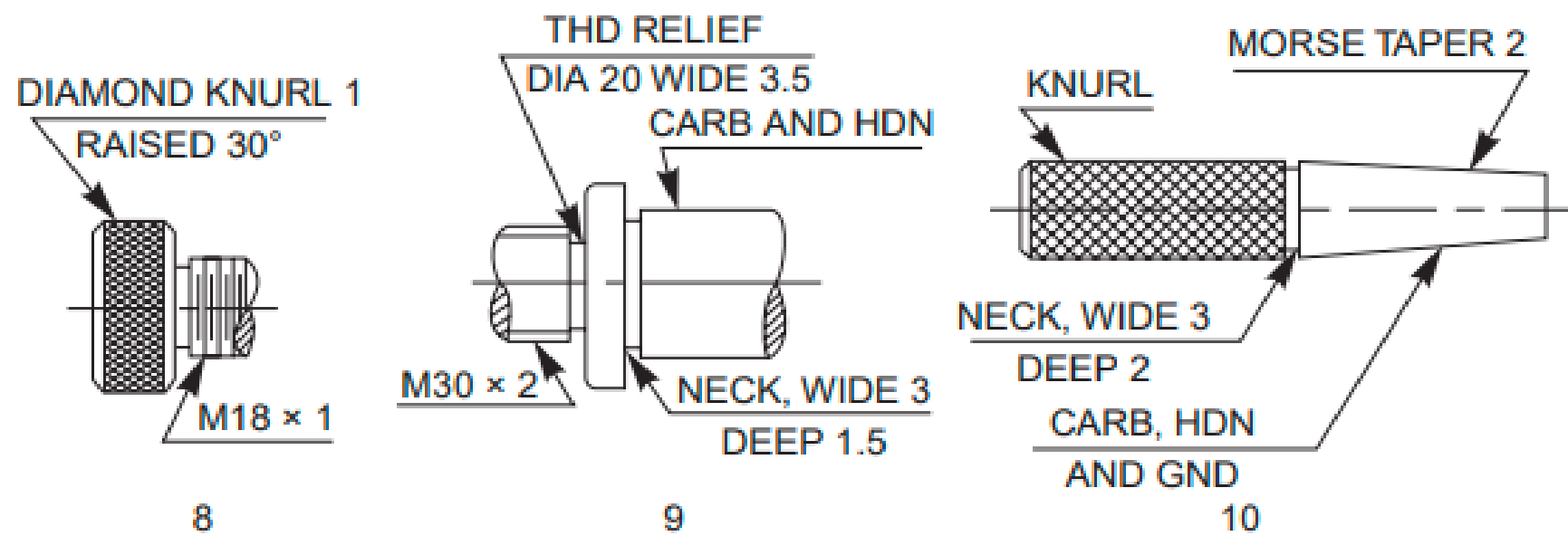


Fig. 2.55 Method of indicating notes (*Contd.*)

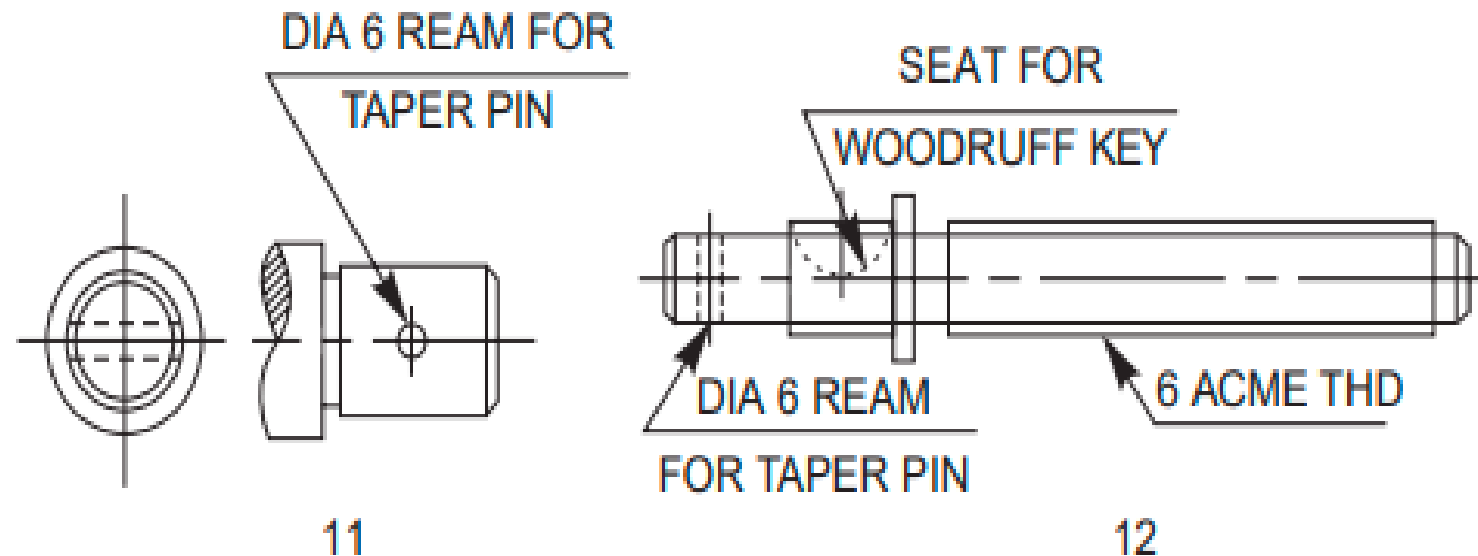
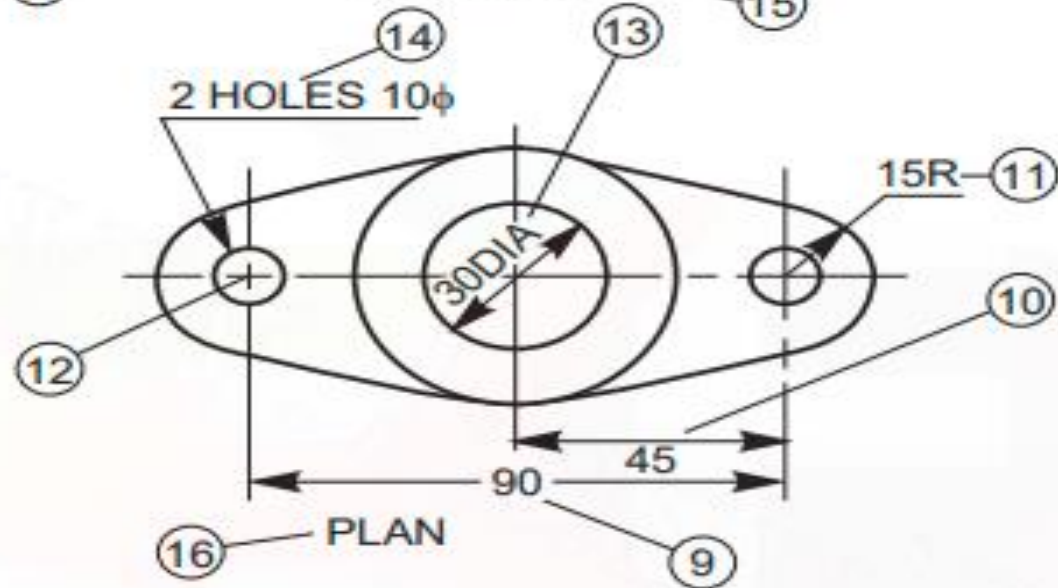
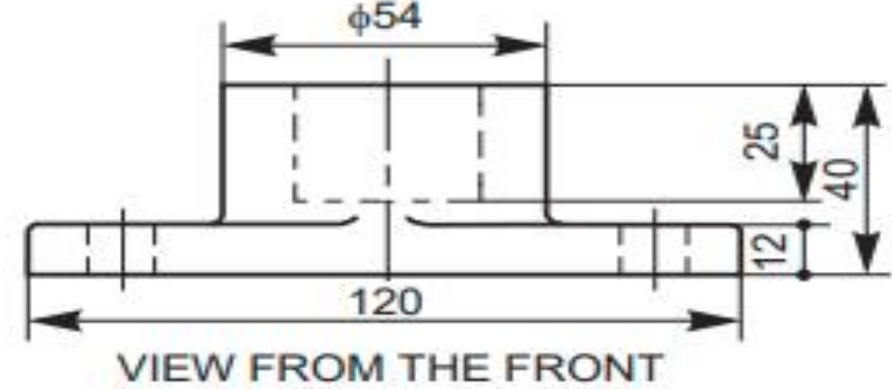
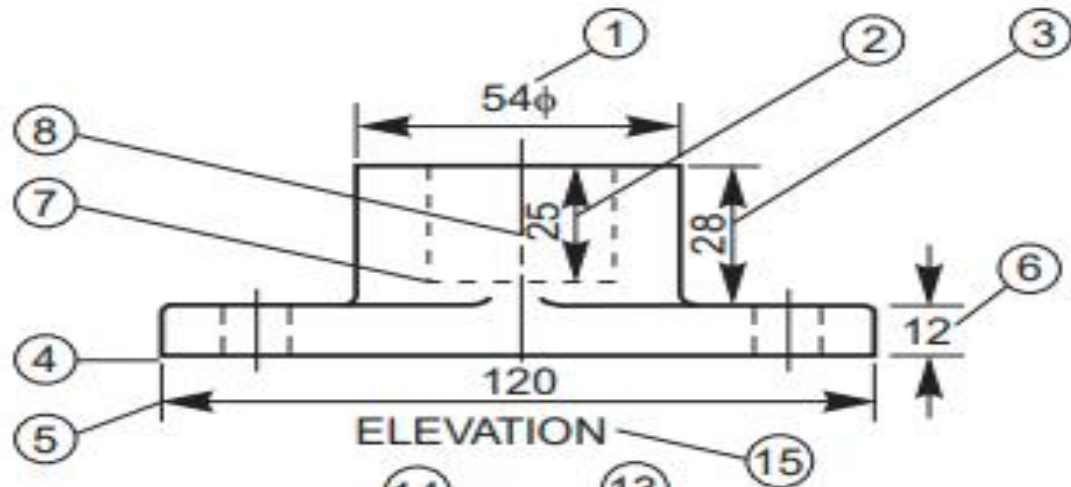


Fig. 2.55 Method of indicating notes

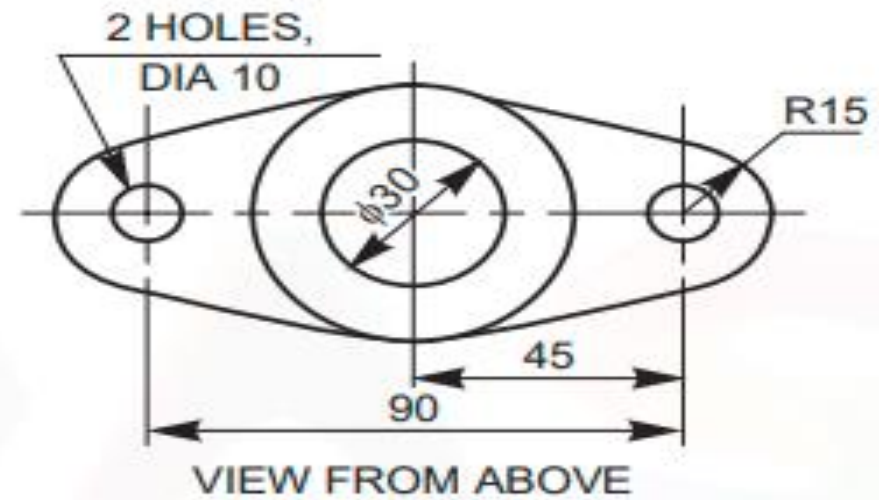
Table 2.8 Meaning of notes given in Fig. 2.55

<i>S.No.</i>	<i>Note</i>	<i>Meaning / Instruction</i>
1.	DIA 25 DEEP 25	Drill a hole of diameter 25 mm, to a depth of 25 mm.
2.	DIA 10 CSK DIA 15	Drill a through hole of diameter 10 mm and countersink to get 15 mm on top.
3.	4 HOLES, DIA 12 C BORE DIA 15 DEEP 8	Drill through hole of ϕ 12 mm, counterbore to a depth of 8 mm, with a ϕ 15 mm, the number of such holes being four.
4.	6 HOLES, EQUI-SP DIA 17 C BORE FOR M 16 SOCKET HD CAP SCR	Drill a through hole of ϕ 17 and counterbore to insert a socket headed cap screw of M 16. Six holes are to be made equi-spaced on the circle.
5.	KEYWAY, WIDE 6 DEEP 3	Cut a key way of 6 mm wide and 3 mm depth.
6.	KEY SEAT, WIDE 10 DEEP 10	Cut a key seat of 10 mm wide and 10 mm deep to the length shown.
7.	U/C, WIDE 6 DEEP 3	Machine an undercut of width 6 mm and depth 3 mm.
8.	(a) DIAMOND KNURL 1 RAISED 30° (b) M 18 $\times$ 1	Make a diamond knurl with 1 mm pitch and end chamfer of 30°. Cut a metric thread of nominal diameter 18 mm and pitch 1 mm.
9.	(a) THD RELIEF, DIA 20 WIDE 3.5 (b) NECK, WIDE 3 DEEP 1.5 (c) CARB AND HDN	Cut a relief for thread with a diameter of 20.8 mm and width 3.5 mm. Turn an undercut of 3 mm width and 1.5 mm depth. Carburise and harden.
10.	(a) CARB, HDN AND GND (b) MORSE TAPER 2	Carburise, harden and grind. Morse taper No. 1 to be obtained.
11.	DIA 6 REAM FOR TAPER PIN	Drill and ream with taper reamer for a diameter of 6 mm to suit the pin specified.
12.	6 ACME THD	Cut an ACME thread of pitch 6 mm.

Practice Examples



a-Incorrect



b-Correct

Fig. 2.56

Violations of some of the principles of drawing are indicated in Fig. 2.56 a. The corrected version of the same as per the BIS, SP-46: 1988 is given in Fig. 2.56 b and the reasons are given below:

1. Dimension should follow the shape symbol (Fig. 2.42).
2. and 3. As far as possible, features should not be used as extension lines for dimensioning.
4. Extension line should touch the feature.
5. Extension line should project beyond the dimension line.
6. Writing the dimension is not as per the aligned system.
7. Hidden lines should meet without a gap (Table 2.5 A).
8. Centre line representation is wrong. Dot should be replaced by a small dash.
9. Horizontal dimension line should not be broken to insert the value of the dimension (Figs. 2.36 to 2.49)

10. Dimension should be placed above the dimension line (Fig. 2.39).
11. Radius symbol should precede the dimension (Fig. 2.42)
12. Centre lines should cross at long dashes (Table 2.5 B).
13. Dimension should be written by symbol (not abbreviation) followed by its value (Fig. 2.42).
14. Note with dimensions should be written in capitals.
15. Elevation is not the correct usage.
16. Usage of the term “plan” is obsolete in graphic language.

Thank you!!!